



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab experiment-24

Experiment 24. Monkey Banana Problem

Aim

To implement the **Monkey Banana Problem** using **Prolog**.

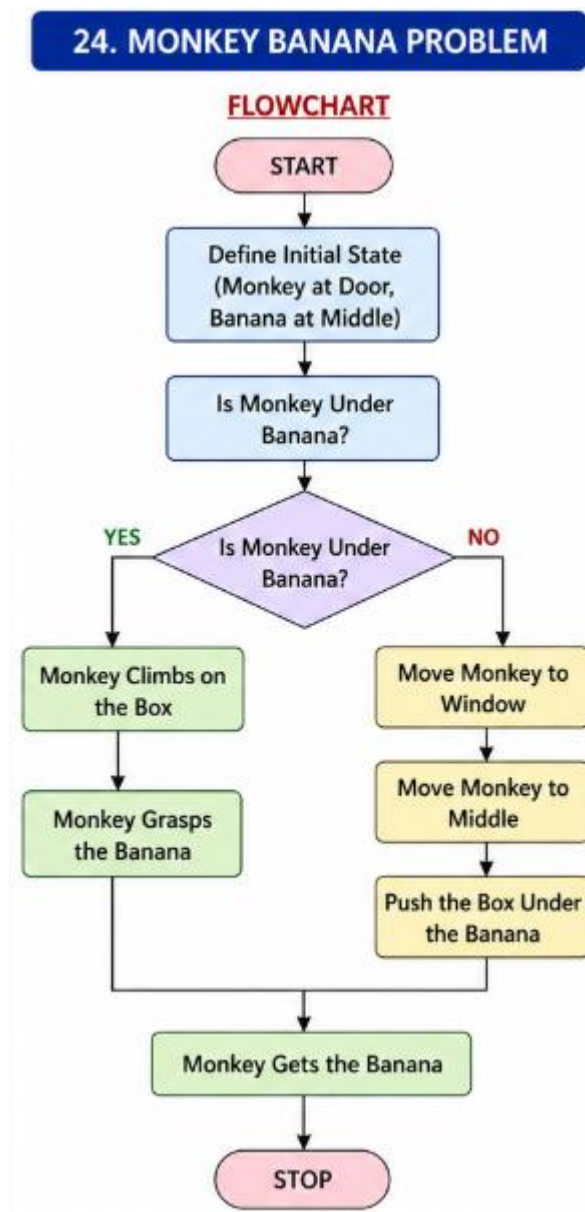
Objectives

- To understand state-space representation.
- To solve a simple planning problem using Prolog.
- To determine the sequence of actions required to get the banana.

Algorithm

1. Define the initial state of the monkey.
2. Define the position of the banana.
3. Check whether the monkey is under the banana.
4. If not, move the monkey.
5. Push the box under the banana.
6. Climb the box.
7. Grasp the banana.

Flowchart



Code

```
move(monkey, door,  
window).
```

```
move(monkey, window,  
middle).
```

```
get_banana :-
```

```
    move(monkey, door,  
window),
```

```
    move(monkey, window,  
middle),
```

```
    write('Monkey moved to  
the middle. '), nl,
```

```
    write('Monkey climbed  
the box. '), nl,
```

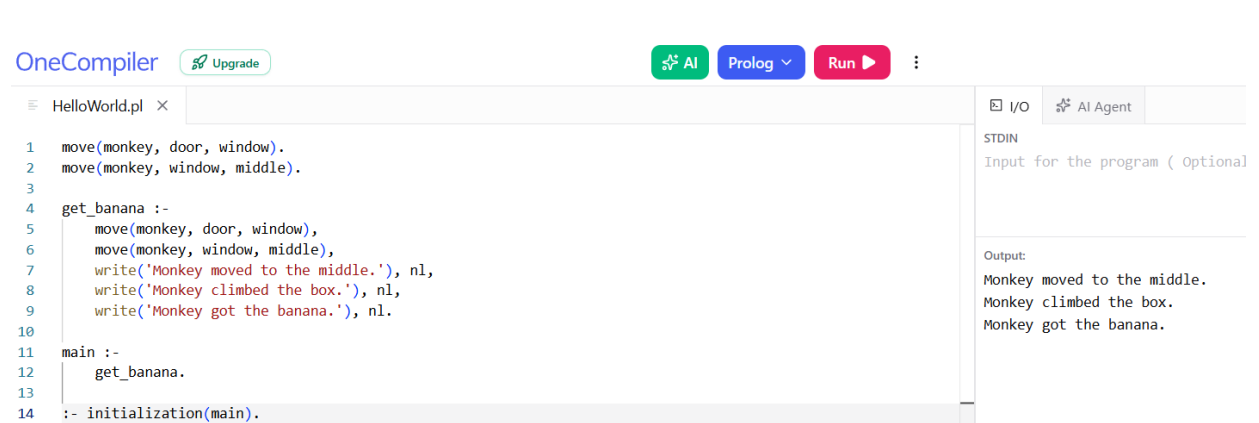
```
    write('Monkey got the  
banana. '), nl.
```

```
main :-
```

```
    get_banana.
```

```
:- initialization(main).
```

Output



The screenshot shows the OneCompiler Prolog IDE. The editor displays a Prolog program named 'HelloWorld.pl' with the following code:

```
1 move(monkey, door, window).
2 move(monkey, window, middle).
3
4 get_banana :-
5     move(monkey, door, window),
6     move(monkey, window, middle),
7     write('Monkey moved to the middle.'), nl,
8     write('Monkey climbed the box.'), nl,
9     write('Monkey got the banana.'), nl.
10
11 main :-
12     get_banana.
13
14 :- initialization(main).
```

The right-hand panel shows the I/O section with the output:

```
Output:
Monkey moved to the middle.
Monkey climbed the box.
Monkey got the banana.
```

d

Result

Monkey moved to the middle.

Monkey climbed the box.

Monkey got the banana.

Conclusion

Thus, the Monkey Banana Problem was successfully implemented using Prolog.